

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 170 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Area Under The Curves — Standard Calculations & Problem Archetypes	95 PYQs	Quantitative Analysis
Very High	Area Under The Curves — Applied Cases, Reactions & Traps	39 PYQs	Applied Topics
Very High	Area Under The Curves — Core Principles & Governing Laws	37 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Area Under The Curves — Standard Calculations & Problem Archetypes	$\int_0^1 (\sqrt{x}-x)dx$, $(y-2)^2=x-1$, $y=x^2$
Area Under The Curves — Applied Cases, Reactions & Traps	$y=3x, 2y=27-3x$, $y= \alpha x-5 - 1-\alpha x +\alpha x^2$, $\{(x,y): x-2y+4 \leq 0, x+2y^2 \leq 0, x+4y^2 \leq 8, y \geq 0\}$
Area Under The Curves — Core Principles & Governing Laws	is equal to _____, $\Rightarrow y=\frac{-4 \pm \sqrt{16+16}}{2}$, $g(x)=\begin{aligned} x^2 \text{ \& } x \geq 0 \\ x \text{ \& } x < 0 \end{aligned}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Area Under The Curves — Standard Calculations & Problem Archetypes (Very High Yield — 95 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Area Under The Curves.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $f:(1,\infty) \rightarrow R$ be a function defined as $f(x)=(x-1/x+1)$. Let $f^{i+1}(x)=f(f^i(x))$, $i=1,2,\dots,25$, where $f^1(x)=f(x)$." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The area of the region $R=(x,y):xy \leq 27, 1 \leq y \leq x^2$ is equal to" [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The area of the region $R=(x,y):xy \leq 8, 1 \leq y \leq x^2, x \geq 0$ is..." [Ans: N/A]

3.2 Area Under The Curves — Applied Cases, Reactions & Traps (Very High Yield — 39 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Area Under The Curves.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x=0, x=1, y^2=x$ and $y=|\alpha x-5|+1-|\alpha x|+|\alpha x^2|$..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (Numerical): "If the area of the region bounded by $16x^2-9y^2=144$ and $8x-3y=24$ is A , then $3(A+6\log_e(3))$ is equal to" [Ans: 24]
- [2026 Morning Shift] Q#3 (MCQ): "The area of the region $(x,y):0 \leq y \leq 6-x, y^2 \geq 4x-3, x \geq 0$ is" [Ans: B]

3.3 Area Under The Curves — Core Principles & Governing Laws (Very High Yield — 37 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Area Under The Curves.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Area Under The Curves.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Area Under The Curves PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#3 (MCQ): "The area of the region $(x,y):y \leq \pi-|x|, y \leq |x \sin x|, y \geq 0$ is:..." [Ans: B]
- [2026 Morning Shift] Q#3 (MCQ): "Let A_1 be the bounded area enclosed by the curves $y=x^2+2, x+y=8$ and y -axis that lies in the first quadrant. Let A_2 be the b..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (Numerical): "Let the area of the region bounded by the curve $y=\max(\sin x, \cos x)$, lines $x=0, x=(3\pi/2)$, and the x -axis be A . Then, $A+A^2$..." [Ans: 12]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x, x \in \mathbb{R}$. Let the maximum value of f on $\mathbb{R}$ be α . If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3, \beta > 0$, is α^2 , then $30\beta^3$ is equal to ____.

Correct Answer: 16

Solution Analysis: $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x \dots (1)$ Now replace x by $\left(\frac{\pi}{2} - x\right)$ in (1). This is a standard NCERT step to get a second equation in the same two terms. $\Rightarrow f\left(\frac{\pi}{2} - x\right) + 3f(x) = \cos x \dots (2)$ Now we solve the two linear equations (1) and (2) to find $f(x)$. Multiply (1) by 3: $3f(x) + 9f\left(\frac{\pi}{2} - x\right) = 3\sin x \dots (3)$ Subtract (2) from (3): $(3f(x) - 3f(x)) + (9 - 1)f\left(\frac{\pi}{2} - x\right) = 3\sin x - \cos x$ $8f\left(\frac{\pi}{2} - x\right) = 3\sin x - \cos x$ Now put $x \rightarrow \left(\frac{\pi}{2} - x\right)$ in this result to get $f(x)$: $8f(x) = 3\cos x - \sin x$ $f(x) = \frac{1}{8}(3\cos x - \sin x)$ To find the maximum value,

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#2)

If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A , then $3(A + 6\log_e(3))$ is equal to _____.

Correct Answer: 24

Solution Analysis: $16x^2 - 9y^2 = 144$ $16x^2 - (8x - 24)^2 = 144$ $16x^2 - 64(x - 3)^2 = 144 \rightarrow x^2 - 4(x - 3)^2 = 9$ $3x^2 - 24x + 45 \rightarrow x^2 - 8x + 15 = 0$ $x = 3, 5$ $\text{Area} = \int_3^5 \sqrt{\frac{16x^2}{9} - 144} - \left(\frac{1}{2} \cdot 2 \cdot \left(\frac{16}{3}\right)\right) dx = \frac{4}{3} \int_3^5 \sqrt{x^2 - 9} - \left(\frac{16}{3}\right) dx = \frac{4}{3} \left(\frac{x}{2} \sqrt{x^2 - 9} - \frac{9}{2} \log_e(x + \sqrt{x^2 - 9}) \right) \Big|_3^5 - \left(\frac{16}{3}\right)(5 - 3) = \frac{4}{3} \left(\frac{5}{2} \cdot 4 - \frac{9}{2} \log_e 9 - \left(\frac{3}{2}\right) \cdot 0 + \frac{9}{2} \log_e 3 - \left(\frac{16}{3}\right) \cdot 2 \right)$ $\text{Area} = 8 - 6\log_e 3 = A \therefore A + 6\log_e 3 = 8 \Rightarrow 3(A + 6\log_e 3) = 24$

Benchmark Problem 3: JEE Main 2026 (Online) 4th April Morning Shift (Q#3)

The area of the region $(x, y): y \leq \pi - |x|, y \leq |x \sin x|, y \geq 0$ is:

- (A) $1 + \frac{\pi^2}{8}$
- (B) $2 + \frac{\pi^2}{4}$ [Correct]
- (C) $\frac{\pi^2}{8} - 1$
- (D) $4 + \frac{\pi^2}{2}$

Correct Answer: B

Solution Analysis: We need the area of the region $(x, y): y \leq \pi - |x|, y \leq |x \sin x|, y \geq 0$. This means the required area is the area under the curve $y = \min(\pi - |x|, |x \sin x|)$ and above the x -axis. So the area is $A = \int \min(\pi - |x|, |x \sin x|) dx$ over the values of x where the region exists. Since $y \geq 0$ and $y \leq \pi - |x|$, we must have $\pi - |x| \geq 0 \Rightarrow |x| \leq \pi$. Hence the region lies in $[-\pi, \pi]$. Also, both functions $\pi - |x|$ and $|x \sin x|$ are even functions, so the region is symmetric about the y -axis. Therefore, $A = 2 \int_0^\pi \min(\pi - x, x \sin x) dx$. Now for $0 \leq x \leq \pi$, we have